# A chart is an argument

TC1037 · Algorithmic Thinking · Session 13 · Plots with `matplotlib`

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Tecnológico de Monterrey · Campus Querétaro · Group 510

# The hour, in order

| When | What | |
|---|---|---|
| **17:35–17:55** | **Quick exam 5** | Lists. Twenty minutes, Respondus and Trinket. **On paper if you are on a Mac** — come get a copy. |
| **After** | **Plots** | Everything you need for Friday's lab and for two of the five functions in your final program. |

Nothing new is examined today that was not on Friday's slides: the accumulator, `sum`/`max`/`min`, `sorted()` against `.sort()`, a row as a list, a table as a list of rows, `tabla[i][j]`.

# Due dates — Friday and Friday

| When | What | What counts as done |
|---|---|---|
| **Fri 13 Nov 23:59** | **Deliverable 2** `entrega2.docx` | Four to six *preguntas detonadoras* about *your* table. In pairs, one file. The exact shape is on the last slides today. |
| **Fri 16:30** | **Lab 7** lists and charts | The same lab as every TC1037 group. Three chart functions plus a menu. Today is the rehearsal. |
| **Rolling** | **Deliverable 1 fixes** | If I sent you a list of fixes, upload the corrected file. **No penalty** for fixing it. |

# Deliverable 1 — what came back

**Twenty-eight of thirty handed something in.** Every file was opened — every sheet, every column, every row.

**Nobody had duplicate rows.** That part the whole group got right.

Each team has a message from me with its own list. Most lists are three lines long.

**The sentence worth hearing:**

**The table and the explanation are two different pieces of work, and most of you did one of them.**

One team wrote the best note in the group — source, link, licence, every column explained — and picked a table with no numbers in it.

Another handed in the cleanest table I have seen in this course and no note at all.

# The four faults — and what each one does to your program

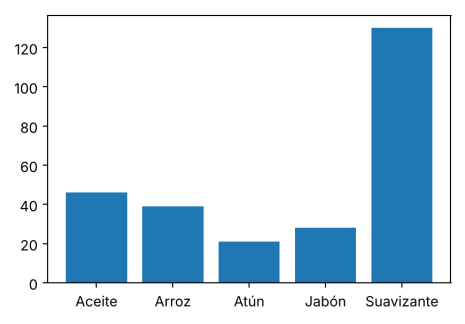
| In the file | In the program, next week |
|---|---|
| Blank rows or columns left in | `len()` counts them. Your mean is divided by the wrong number. **No error message.** |
| A **constant** column — the same value in all 40 rows | Nothing to group by, and its average is that value. It is not data, it is a label. |
| An **identifier** used as a variable — receipt number, time stamp, URL, address | Unique in every row. You cannot count categories in it and its mean means nothing. |
| Numbers stored as **text** | `"10" < "9"` is `True`. Comparisons go alphabetical and the maximum is wrong. |

# Your first chart — five lines

```
import matplotlib.pyplot as plt

productos = ["Aceite", "Arroz", "
    Atun",
              "Jabon", "Suavizante"
    ]
precios   = [46, 39, 21, 28, 130]

plt.bar(productos, precios)
plt.show()
```




**And it says nothing.**

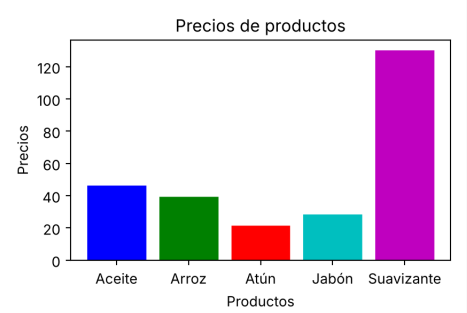
Two lists of the **same length**. The first is what you read on the x axis, the second is how tall each bar is. Position 0 with position 0.

# The same chart, with the argument put back in

```
plt.title("Precios de productos")
plt.xlabel("Productos")
plt.ylabel("Precios")

barras = plt.bar(productos,
    precios)
barras[0].set_color('b')
barras[1].set_color('g')
barras[2].set_color('r')
barras[3].set_color('c')
barras[4].set_color('m')

plt.show()
```

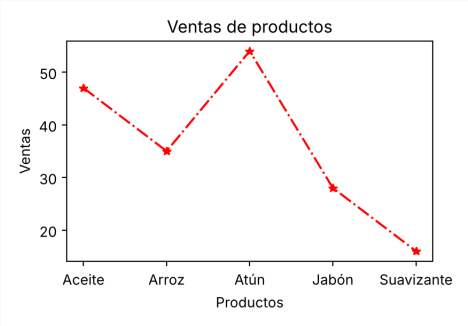



`plt.bar` **returns** the bars. `barras` is a list — one entry per bar — so `barras[0]` is the first bar, exactly like any list you wrote on Friday.

# The line chart — and the three-character style string

```
ventas = [47, 35, 54, 28, 16]

plt.title("Ventas de productos")
plt.xlabel("Productos")
plt.ylabel("Ventas")
plt.plot(productos, ventas, "r*-."
    )
plt.show()
```




`"r*-."` is three decisions in one string:
`r` the colour · `*` the marker at each point · `-.` the line style.

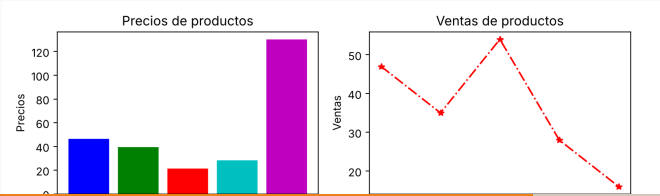
Change it and nothing breaks.

# Two charts, one figure — subplot

```
plt.subplot(121)                          # 1 row, 2 columns, drawing in panel 1
plt.title("Precios de productos")
plt.xlabel("Productos"); plt.ylabel("Precios")
plt.bar(productos, precios)

plt.subplot(122)                          # same grid, now panel 2
plt.title("Ventas de productos")
plt.xlabel("Productos"); plt.ylabel("Ventas")
plt.plot(productos, ventas, "r*-.")

plt.show()                                # ONE show, at the very end
```



# The shape of the program — you already know it

```
def menu():
    print("1. Grafica de precios")
    print("2. Grafica de ventas")
    print("3. Grafica de precios y ventas")
    print("4. Salir")
def main():
    productos = [...]         # the three lists
    precios   = [...]
    ventas    = [...]
    continua = True
    while continua == True:
        menu()
        opcion = int(input("Opcion: "))
        if opcion == 1:
            precios_productos(productos,
    precios)
        elif opcion == 4:
            continua = False
main()
```

**Nothing here is new.**

Functions — week 2. `if/elif/else` — week 2. `while` — week 3. Lists — Friday. The only new line is `plt.bar`.

**Every `input()` lives in `main`.**

The chart functions receive lists and draw. They never ask the user anything.

This is also the skeleton of your **final program** — same menu, same loop, with `estadistica_descriptiva` added.

# Names: copy them exactly

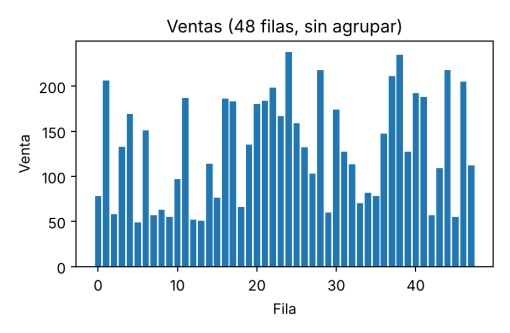
The lab and the project both use **Spanish** function names.

| | |
|---|---|
| **Friday's lab** | `precios_productos` · `ventas_productos` · `precios_ventas` · `menu` · `main` |
| **Your final program** | `estadistica_descriptiva` · `grafica1` · `grafica2` · `menu` · `main` |

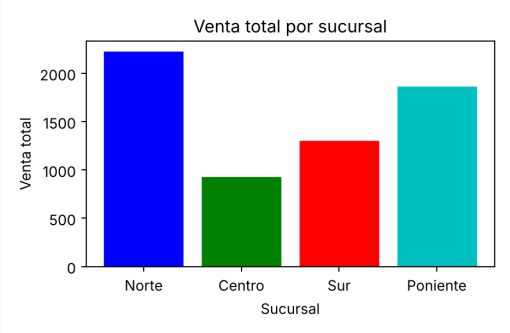
**A name that is nearly right is wrong.** `graficar1`, `Grafica1` and `grafica_1` are three different functions, and none of them is the one being looked for.

File for Friday: `G_Matricula.py` — your own matrícula, a `.py` file, uploaded to the lab activity.

# The trap: you cannot plot your table




Forty-eight rows, one bar each. True, complete — and it answers nothing.




The same forty-eight rows. **Group first, then draw.**

**matplotlib needs two short lists. Your table has forty rows. The work in between is the loop you wrote on Friday.**

# From a table to two lists — Friday's loop, one more time

```
tabla = [["Norte", 120], ["Centro", 80], ["Sur", 95], ["Norte", 60], ["Sur", 40]]

nombres = []        # the x axis: one entry per branch
totales = []        # the y axis: the total for that branch

for fila in tabla:
    sucursal = fila[0]
    venta    = fila[1]
    if sucursal in nombres:
        i = nombres.index(sucursal)
        totales[i] = totales[i] + venta        # the accumulator, again
    else:
        nombres.append(sucursal)
        totales.append(venta)

plt.bar(nombres, totales)
```

`nombres` → `['Norte', 'Centro', 'Sur']`  `totales` → `[180, 80, 135]`

# Deliverable 2 — due Friday. The exact shape.

**Four to six preguntas detonadoras about your own table.** One Word file per team.

Each question must carry three things with it:

1. **The question** a manager would actually ask. *Which branch sells most?* Not *what is in the data?*
2. **The column or columns** that answer it — by their real names, as they are spelled in your file.
3. **The statistic or the chart** that answers it. *Total by branch, bar chart.*

And at the end: **who owns which**. Each of you takes two or three, written down.

Marks: 50 for questions answerable *from your columns* · 30 for naming the column and the statistic or chart · 20 for the ownership.

# One worked example — and one that fails

| Question | Columns | Statistic / chart |
|---|---|---|
| Which branch sells most? | `Sucursal`, `Venta` | Total by branch · bar chart |
| Which product is returned most often? | `Producto`, `Devuelto` | Count by product · bar chart |
| Did sales grow from January to June? | `Mes`, `Venta` | Total by month · line chart |
| Is the expensive line also the best seller? | `Precio`, `Venta` | Two panels, one figure |
| Are our customers satisfied? | — | No column holds this. |

The fourth one is `grafica2` in your final program, already chosen. The red one is a good business question and a bad deliverable-2 question.

# Asesorías — the offer is open, and it is working

**I cancelled sessions two weeks ago without telling anyone.** I apologised for that on Friday and the schedule has been clear since.

**Slots are being taken.** Those who came brought a file and left with it working. That is the whole mechanism — there is no other trick to this course.

Booking link: the Canvas announcement. One click, pick a time.

**Bring one specific thing:**

- your table, if I sent you fixes and you are not sure what they mean;
- your four questions, before Friday, for ten minutes of sanity checking;
- the chart that will not draw;
- your midterm paper, if the mark still does not make sense to you.

**Fifteen minutes on a real problem beats an hour of nodding.**

# Before Friday

1. **Fix your table** if I sent you a list. It is three lines of work and there is no penalty.
2. **Write the four to six questions** — question, columns, statistic or chart, who owns it. `entrega2.docx`, Friday 23:59.
3. **Run today's five lines** on your own two lists before the lab, so Friday is practice and not a first attempt.
4. **Book an asesoría** if any of the three above is not obvious by Thursday.

**A chart that needs you standing next to it is not finished.**

Title, both axis labels, and a legend when there is more than one series. Every time.